DO NOW — RECALL

# byVal, byRef and modular programming

Name: _____ Class: _____ Date: _____

**1. What does return do in a C# function?**

**2. Can a method read a local variable declared inside another method?**

**3. What is a[0] for int[] a={9,6,3}?**

**4. If Double(n) returns 2*n, find Double(Double(3)).**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Finishes the invocation and returns a result to its caller.

### 2. Model answer

Not directly. It is outside the variable's scope.

### 3. Model answer

9.

### 4. Model answer

Double(3)=6; Double(6)=12.

### Vocabulary — know these terms

by value — The subroutine receives a copy of the value.

by reference — The parameter refers to the caller's variable.

modularity — Divide a program into smaller testable subroutines.

ref — C# keyword enabling reference parameter passing.

side effect — A change to external program state.

NEW LEARNING

# Read, annotate and explain

### CALLER

The caller declares marks = 10.

### ALIAS

ref makes score refer to the caller's marks rather than an independent copy.

### AFTER

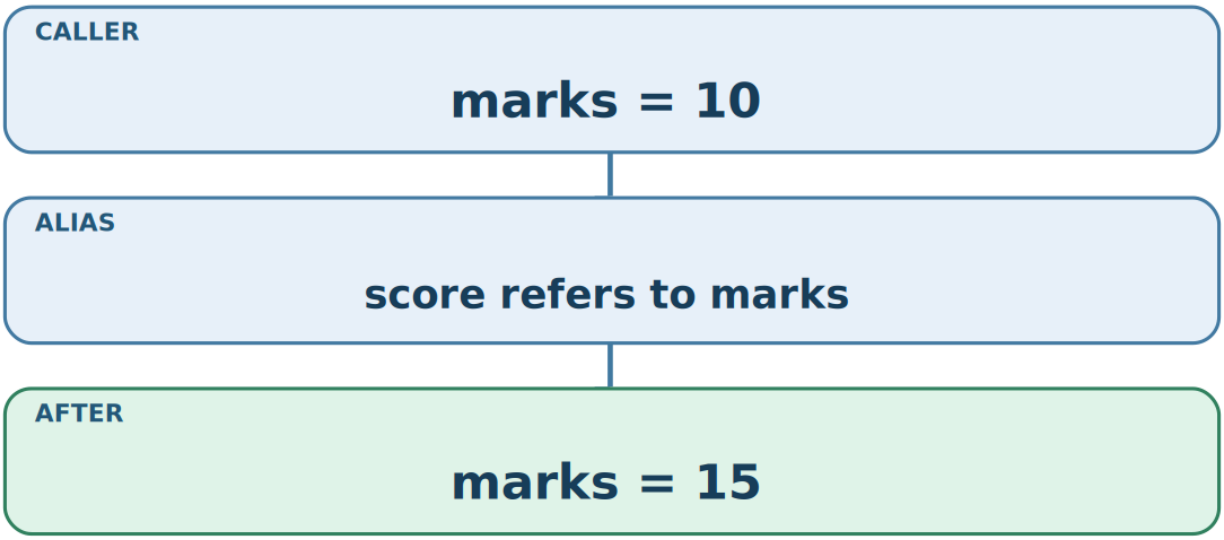
Changing score therefore changes marks, which becomes 15.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Predict what happens**




CALLER: marks = 10; ALIAS: score refers to marks; AFTER: marks = 15

I DO — WORKED EXAMPLE

# Follow the worked method

### A function receives a copy of x=7 and adds 4. What is x afterwards?

### Predict first

A new parameter value receives a copy of the integer 7.

### Model the result

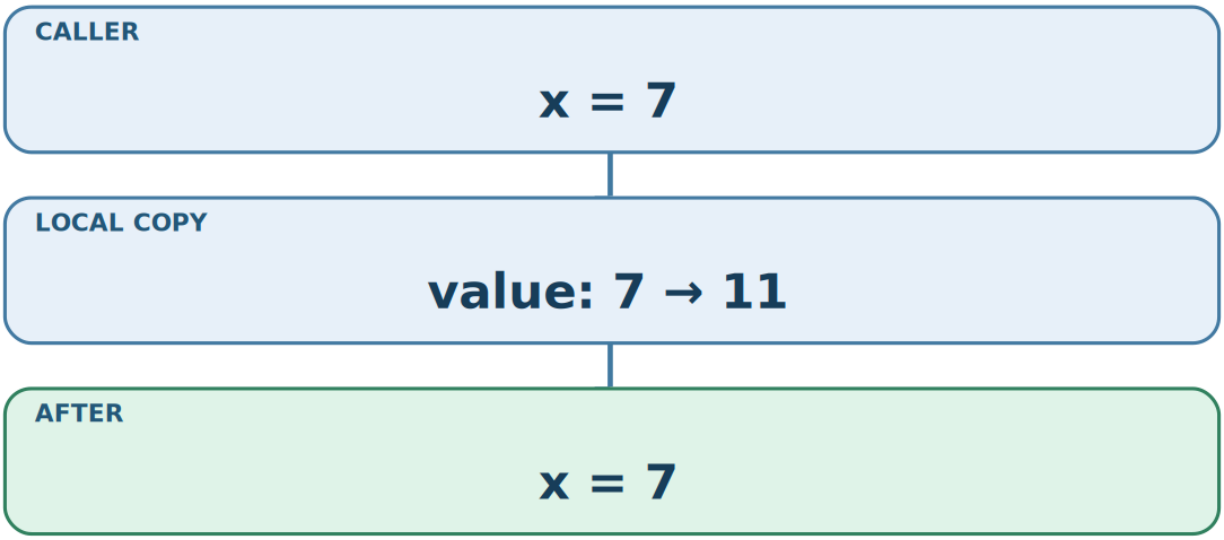
value becomes 11 inside AddFour, but x remains 7 after the call.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**A function receives a copy of x=7 and adds 4. What is x afterwards?**

A new parameter value receives a copy of the integer 7.




CALLER: x = 7; LOCAL COPY: value: 7 → 11; AFTER: x = 7

WE DO — GUIDED PRACTICE

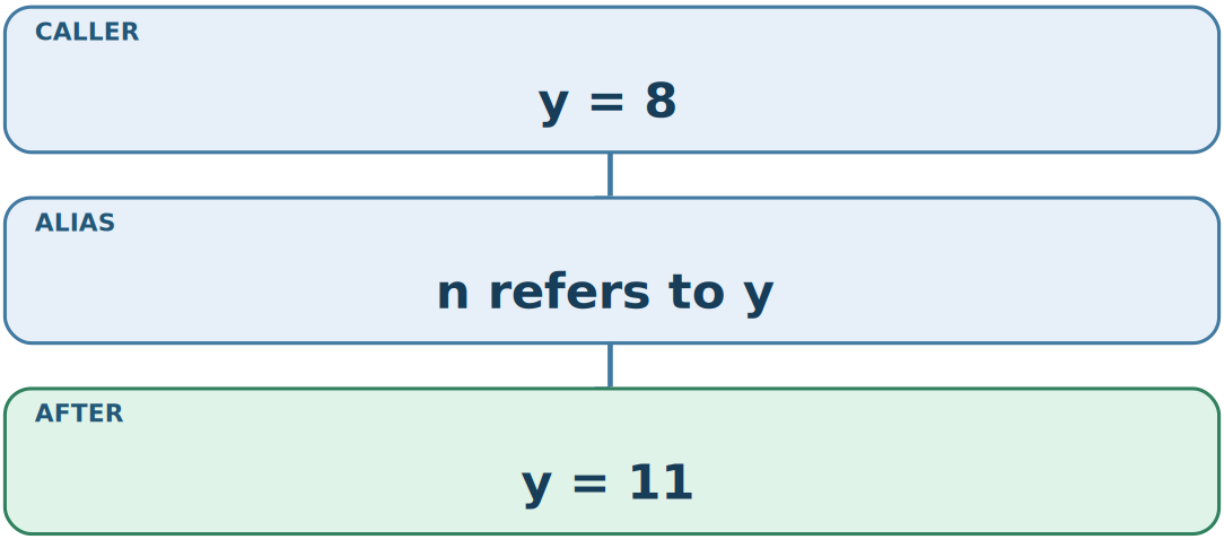
# Try → Discuss → Improve

# We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**A procedure adds 3 to a ref parameter. Trace y=8 through the call.**

y starts at 8 and is passed using ref.




CALLER: y = 8; ALIAS: n refers to y; AFTER: y = 11

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

### Guided 1: Predict first y starts at 8 and is passed using ref.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

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YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

A method takes int number by value and adds 2. Caller x=5. What is x after the call?

### Question 2

Write a C# procedure that doubles an integer passed by reference.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Explain when returning a result is preferable to changing a ref parameter.

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Changing a by-value int parameter always changes the caller's int.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

## GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

### Guided 1 — original CHECK

Model the result

n refers to y, so adding 3 makes y = 11.

IMPROVE: Explain one correction to your own reasoning.